

DPP NO.-5

- Q.1 There are m points on a straight line AB & n points on the line AC none of them being the point A. Triangles are formed with these points as vertices, when
- (i) A is excluded (ii) A is included. The ratio of number of triangles in the two cases is:
- (A) $\frac{m+n-2}{m+n}$ (B) $\frac{m+n-2}{m+n-1}$ (C) $\frac{m+n-2}{m+n+2}$ (D) $\frac{m(n-1)}{(m+1)(n+1)}$
- Q.2 Number of ways in which 7 green bottles and 8 blue bottles can be arranged in a row if exactly 1 pair of green bottles is side by side, is (Assume all bottles to be alike except for the colour).
- (A) 84 (B) 360 (C) 504 (D) 84
- Q.3 In a certain algebraical exercise book there are 4 examples on arithmetical progressions, 5 examples on permutation – combination and 6 examples on binomial theorem . Number of ways a teacher can select for his pupils atleast one but not more than 2 examples from each of these sets, is _____ .
- Q.4 The kindergarten teacher has 25 kids in her class . She takes 5 of them at a time, to zoological garden as often as she can, without taking the same 5 kids more than once. Find the number of visits, the teacher makes to the garden and also the number of of visits every kid makes.
- Q.5 There are n persons and m monkeys ($m > n$). Number of ways in which each person may become the owner of one monkey is
- (A) n^m (B) m^n (C) mP_n (D) mn
- Q.6 Seven different coins are to be divided amongst three persons . If no two of the persons receive the same number of coins but each receives atleast one coin & none is left over, then the number of ways in which the division may be made is :
- (A) 420 (B) 630 (C) 710 (D) none
- Q.7 Let there be 9 fixed points on the circumference of a circle . Each of these points is joined to every one of the remaining 8 points by a straight line and the points are so positioned on the circumference that atmost 2 straight lines meet in any interior point of the circle . The number of such interior intersection points is :
- (A) 126 (B) 351 (C) 756 (D) none of these
- Q.8 The number of 5 digit numbers such that the sum of their digits is even is :
- (A) 50000 (B) 45000 (C) 60000 (D) none
- Q.9 A forecast is to be made of the results of five cricket matches, each of which can be win, a draw or a loss for Indian team. Find
- (i) the number of different possible forecasts
- (ii) the number of forecasts containing 0, 1, 2, 3, 4 and 5 errors respectively

- Q.10 The number of ways in which 8 distinguishable apples can be distributed among 3 boys such that every boy should get atleast 1 apple & atmost 4 apples is $K \cdot {}^7P_3$ where K has the value equal to
 (A) 14 (B) 66 (C) 44 (D) 22
- Q.11 A women has 11 close friends. Find the number of ways in which she can invite 5 of them to dinner, if two particular of them are not on speaking terms & will not attend together.
- Q.12 A rack has 5 different pairs of shoes. The number of ways in which 4 shoes can be chosen from it so that there will be no complete pair is
 (A) 1920 (B) 200 (C) 110 (D) 80

Paragraph for question nos. 13 to 15

Consider the word $W = \text{MISSISSIPPI}$

- Q.13 If N denotes the number of different selections of 5 letters from the word $W = \text{MISSISSIPPI}$ then N belongs to the set
 (A) {15, 16, 17, 18, 19} (B) {20, 21, 22, 23, 24}
 (C) {25, 26, 27, 28, 29} (D) {30, 31, 32, 33, 34}
- Q.14 Number of ways in which the letters of the word W can be arranged if atleast one vowel is separated from rest of the vowels
 (A) $\frac{8! \cdot 161}{4! \cdot 4! \cdot 2!}$ (B) $\frac{8! \cdot 161}{4 \cdot 4! \cdot 2!}$ (C) $\frac{8! \cdot 161}{4! \cdot 2!}$ (D) $\frac{8!}{4! \cdot 2!} \cdot \frac{165}{4!}$
- Q.15 If the number of arrangements of the letters of the word W if all the S's and P's are separated is $(K) \left(\frac{10!}{4! \cdot 4!} \right)$ then K equals
 (A) $\frac{6}{5}$ (B) 1 (C) $\frac{4}{3}$ (D) $\frac{3}{2}$
- Q.16 In how many different ways a grandfather along with two of his grandsons and four grand daughters can be seated in a line for a photograph so that he is always in the middle and the two grandsons are never adjacent to each other.